

EP 10 Operation of a Cathode Ray Oscilloscope

1. Objective

1. To understand the principle of a cathode ray oscilloscope (CRO).
2. To learn how to measure an alternating current (A.C) or direct current (D.C) voltage using a CRO.
3. To learn how to measure an unknown frequency using a CRO.

2. Principle

The CRO is a fundamental and versatile instrument in physics laboratories. Conventionally, it is an analog system (some modern high-performance CROs digitize and process signals digitally) that provides two-dimensional displays of analog signals. It consists of three core components: an **electron gun**, a **fluorescent screen**, and two pairs of **deflection plates** (X-plates for horizontal deflection, Y-plates for vertical deflection); additional key components include trigger units.

When voltage signals V_x (X-plates) and V_y (Y-plates) are applied, the horizontal deflection x and vertical deflection y are proportional to V_x and V_y (or quantities converted to voltages). The voltage fed to X-plates can be:

- An external V_x : The CRO displays V_y vs V_x .
- A time base voltage: Generated by a time base generator (rises steadily then drops rapidly), making the screen spot sweep left-to-right and reset. This displays V_y vs time t .

A triggering pulse (from the Y-signal or input signal via the trigger unit) synchronizes the time base frequency with the input signal for stable waveform display.

3. Procedure

3.1 Adjust the Device

3.1.1 Initial Setting Up (for all models of CRO)

Before connecting the power cord, confirm the AC line voltage selector matches the local supply. Set switches/controls as follows:

POWER (OFF), INTENSITY/FOCUS (mid), VERT MODE (CH1), VOLTS/DIV (500 mV/DIV), AC-GND-DC (GND), SWEEP MODE (AUTO), TIME/DIV (0.5 ms/DIV), and all VARIABLE knobs to "CAL'D" (fully clockwise).

3.1.2 Displaying an A.C Waveform (Using a Signal Generator)

1. Connect the power cord, turn on POWER, and wait ~20s for a trace (recheck settings if no trace after 60s).
2. Adjust INTENSITY and FOCUS for appropriate brightness and sharpness.
3. Align the trace with the graticule's horizontal center using CH1 POSITION.
4. Connect the probe to CH1 INPUT, re-adjust FOCUS.
5. Set the signal generator to SINE OUTPUT (100 Hz), connect to the CRO, and switch AC-GND-DC to AC.
6. Adjust VOLTS/DIV (appropriate amplitude) and TIME/DIV (4–5 peaks) to display the waveform; align with the graticule using $\uparrow/\leftrightarrow$ POSITION. Record SWEEP TIME/DIV.
7. Repeat Step 6 for signal generator frequencies of 1 kHz, 10 kHz, 100 kHz; record SWEEP TIME/DIV.

3.2 Measuring Voltage

3.2.1 Measuring the virtual Value of an A.C Voltage

Set the signal generator to 1 kHz, adjust VOLTS/DIV and TIME/DIV. Measure V_{p-p} (peak-to-peak voltage) using:

$$V_{p-p} = H \times \text{VOLTS/DIV} \times \text{Probe Attenuation Ratio}$$

(Where H = graticule divisions between peak and trough.) Calculate RMS value:

$$U = \frac{\sqrt{2}}{4} \times V_{p-p}$$

3.2.2 Measuring a D.C Voltage

Connect a D.C power supply to CH1 input-earth (any polarity), set AC-GND-DC to DC. Adjust the power supply to 1.5 V, measure the trace shift (cm), and convert to volts using the current VOLTS/DIV setting. Repeat with reversed polarity.

3.3 Measuring Unknown Frequencies

3.3.1 Single-Channel Operation

Use the TIME/DIV scale to measure the signal period T (horizontal distance of 1 full cycle/cm). Calculate frequency $f = 1/T$. Set the signal generator to 50 Hz (nominal) and measure its "true" frequency.

3.3.2 Using Lissajous Figures

Connect the unknown sine signal to CH1 and the signal generator's SINE OUTPUT to CH2. Adjust the unknown frequency until Lissajous figures appear. Use the frequency relationship

$$\frac{f_x}{f_y} = \frac{N_y}{N_x}$$

(N_x = intersections with horizontal line, N_y = intersections with vertical line) to calculate the unknown frequency.

4. Data Recording and Processing

4.1 Data Recording

(A) Table 1: SWEEP TIME/DIV vs. Frequency

	100Hz	1KHz	10KHz	100KHz
SWEEP TIME/DIV(ms)				

(B)

(i)

$V_{p-p} = , U_{A.C} =$

(ii)

divisions shifted(0.5V/cm): $d_1 =$
 $U_{D.C} =$

divisions shifted(1V/cm): $d_2 =$
divisions shifted(0.2V/cm): $d_3 =$

(C) Table 2

Nominal freq. (Hz)	Sweep time(s/div)	Wavelength (cm)	Period (s)	"True" freq. (Hz)	percentage error (%)
50					

Table 3

TABLE 3

N_X/N_Y	The known frequencies	The unknown frequencies
1:1		
1:2		
1:3		
2:3		

4.2 Data Analysis

4.2.1 Data Validity Verification

4.2.2 Quality of Linear Fitting

4.2.3 Result Consistency Analysis

4.2.4 Error Source Validation

5. Error Analysis

5.1 Systematic Errors

- **Uncalibrated VARIABLE Controls:** VOLTS/DIV and TIME/DIV scales are only valid when VARIABLE is fully clockwise ("CAL'D"); deviation leads to inaccurate voltage/period measurements.
- **AC Coupling for D.C Measurement:** AC mode inserts a capacitor that blocks D.C components, causing incorrect trace shift readings.
- **Poor Trigger Synchronization:** Improper LEVEL settings make waveforms drift, leading to wrong V_{p-p} or period readings.

5.2 Random Errors

- **Parallax Error:** Reading the trace at an angle miscounts graticule divisions.
- **Probe Contact Instability:** Loose connections distort waveforms or fluctuate amplitudes.
- **Signal Generator Nominal Deviation:** Displayed frequency differs from actual output, affecting frequency comparison.

6. Questions

Question 1

The VARIABLE controls of the VOLTS/DIV scale and TIME/DIV scale must be fully clockwise when you measure the voltage or frequency, why?

Answer: The VOLTS/DIV and TIME/DIV scales are calibrated by the manufacturer to provide accurate values (e.g., 500 mV per division, 0.5 ms per division) only when the VARIABLE controls are in the "CAL'D" (fully clockwise) position. If the VARIABLE controls are not fully clockwise, the actual voltage per division or time per division deviates from the marked scale values. This results in inaccurate calculations of voltage (via V_{p-p}) or frequency (via period T).

Question 2

What happens if you use AC coupling instead of DC coupling when measuring a D.C voltage?

Answer: AC coupling inserts a capacitor in series with the Y-input terminal of the CRO. Capacitors can only pass A.C signals and block D.C signals—once the capacitor charges to the D.C voltage of the input, no further current flows, and the trace does not shift from its initial position. As a result, the D.C voltage cannot be measured, and the reading will be near zero or unstable.